TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

3D Fabric and Apparel Simulation System

The 2D pattern of Experiment 3, sewn together in the computer and draped on a virtual body: how cloth physics, avatar measurements and fit maps let a factory approve fit and styling before a single physical sample is cut.

TE-3208 · SOFTWARE AND AI IN TEXTILE MANUFACTURING SESSIONAL · EXPERIMENT 4 · LEVEL 3 TERM 2

1. Aim

To assemble a garment in a 3D simulation system — importing the 2D pattern, placing and "sewing" the panels on an avatar, assigning measured fabric properties, running the drape simulation — and to evaluate the result with ease and tension maps.

2. Concepts you need

- **The avatar:** a parametric virtual body built from a measurement set (height, chest, waist, hip …) or imported from a 3D body scan. The avatar is the fitting model that never sleeps: every size in the range can have its own avatar, so grading is checked on bodies, not just on paper.
- **Virtual sewing:** the flat panels are positioned around the avatar and their edges are joined with seam lines defined on screen (front side seam to back side seam, sleeve into armhole). The simulator then lets gravity and contact settle the cloth onto the body — the digital equivalent of the sample hand's first fitting.
- **Fabric physics:** cloth is simulated as a mesh of particles joined by virtual springs. Its behaviour comes from measured properties: **bending stiffness** (how sharply it folds), **shear** (how it skews on the bias), **stretch** (warp/weft extension), **friction** (against body and itself), **thickness** and **weight (GSM)**. Two fabrics of the same GSM but different bending stiffness drape quite differently — the numbers, not the name of the fabric, drive the result.
- **Mesh resolution:** the particle spacing (e.g. 20 mm standard, 10 mm fine) sets both realism and computing cost. Halving the spacing roughly quadruples the particle count over a panel — folds become finer, the solve becomes slower.
- **Fit maps:** after the drape settles, the system colours the garment by **ease** (gap between cloth and body), **tension/strain** (where cloth is stretched) and **pressure** (where it presses the body). A tight

armhole or a dragging hem is visible numerically before any fabric is cut.

- **What it replaces — and what it does not:** 3D approval can eliminate one or more rounds of physical proto samples and courier charges, and is now accepted by many buyers for styling and fit pre-approval. Hand-feel, drape in motion and final confirmation still need cloth: simulation narrows the physical sampling, it does not abolish it.

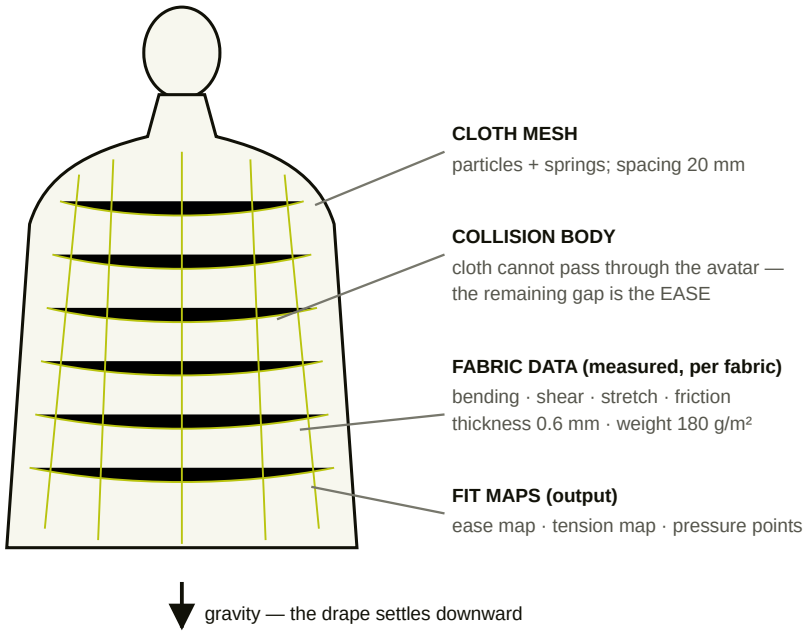


Figure 1 — The elements of a drape simulation: a parametric avatar (collision body), the garment as a particle mesh, measured fabric properties, and gravity. Fit is judged from the maps the settled simulation produces.

3. Lab procedure

1. Import the 2D pattern file of the basic shirt/bodice from Experiment 3 (all pieces, base size M).
2. Create the avatar from the size chart: height 172 cm, chest 107 cm, waist 82 cm, hip 102 cm. Save it as the size-M fitting avatar.
3. Arrange the panels around the avatar (front, back, sleeves) in their wearing positions, then define the seam lines joining them.
4. Assign the fabric: woven cotton poplin, GSM 180 g/m², thickness 0.6 mm, with the bending/shear/stretch values provided in the lab file (or measured values if a fabric property kit is available).
5. Run the simulation at 20 mm particle spacing; wait for the drape to settle (the system reports when movement stops).
6. Read the fit maps: record the wearing ease at chest, waist and hip from the ease map, and note every red (high-tension) zone from the tension map.

7. Make one correction the maps demand — e.g. add 1 cm across the back if the back tension is high — re-run, and record the change in the maps. Export front/side/back views for the report.

4. Worked example — ease readings and mesh cost

Wearing ease is garment measurement minus body measurement at the same level. The settled simulation reports garment circumferences; against the avatar of step 2:

LEVEL	BODY (AVATAR)	GARMENT (SIMULATED)	WEARING EASE	READING
Chest	107 cm	112 cm	**+5 cm**	Normal for a regular-fit shirt
Waist	82 cm	88 cm	**+6 cm**	Comfortable; no tension expected
Hip	102 cm	110 cm	**+8 cm**	Hem will stand away slightly — acceptable

A negative ease at any level in a woven (non-stretch) fabric would show as a red tension zone and a garment that cannot close — the simulation's way of failing a pattern before cloth does.

Mesh cost: a front panel of 60 cm × 50 cm at 20 mm particle spacing carries (60÷2) × (50÷2) = 30 × 25 = **750 particles**. Re-running at the fine 10 mm spacing gives (60÷1) × (50÷1) = 60 × 50 = **3,000 particles** — four times the elements, and a solve several times slower, in exchange for finer folds. Rule of the lab: draft at 20 mm, present at 10 mm.

5. Practice tasks

1. A simulated trouser waistband measures 84 cm on an avatar whose waist is 86 cm. Compute the ease, and state what the tension map will show and what pattern action follows for a woven fabric.
2. Two fabrics both weigh 180 g/m². Fabric A is a stiff canvas, fabric B a soft viscose. Explain why entering only the GSM into the simulator cannot distinguish their drapes, and name the properties that must differ.
3. For a panel 80 cm × 40 cm, compute the particle counts at 20 mm and at 10 mm spacing, and the ratio between them.

Answer key

1. Ease = 84 − 86 = **−2 cm** (negative ease). In a woven fabric the band cannot stretch to close: the tension map will show a continuous high-tension (red) band at the waist and the closure will gap or

fail. Pattern action: add at least 2 cm plus wearing ease (typically +2 to +3 cm at a trouser waist) to the waistband and the adjoining panels, then re-simulate.

2. Drape is governed by how the cloth bends, shears and stretches, not by weight alone. Canvas has high **bending stiffness** and low shear — it stands away in broad, stiff folds; viscose has low bending stiffness and easy shear — it collapses into many fine folds close to the body. GSM (and thickness) must be entered, but the distinguishing inputs are **bending stiffness, shear stiffness and stretch** (with friction affecting how it slides on the body).
3. At 20 mm: (80÷2) × (40÷2) = 40 × 20 = **800 particles**. At 10 mm: (80÷1) × (40÷1) = 80 × 40 = **3,200 particles**. Ratio = 3,200 ÷ 800 = **4×** — halving the spacing quadruples the mesh, as in the worked example.
